Benefits of Open Source software and tools

- There are many open source softwares. so, we can select and use any software that suits our needs.
- The complete options of the software can be used without any cost and restrictions.
- We can share our ideas with the team, write the required code and share it with many.
- As we can identify the programming techniques of group members, we can learn many ideas and make our program writing skills more efficient.
- The coding in open source softwares are being groomed by many enthusiastical members of the group. So if we report problems that we have in the program they are quickly mended by the group's effort.
- As we can make changes to the open source softwares, we can add the most required features in the software
- Many open source software are very user friendly.

Like benefits Open source Software also have Some Problems Like Difficult to work for beginners, Exchange of files to other softwares, Some times Lack of Responsibility, service and problems related to hardware compatibility.

Example of open source Application software

NS2 , OPEN NMS, Ubuntu , MySQL, PDF Creator, Open Office, 7zip GNUCASH, GIMP, BLENDER, AUDACITY, VLC, MOZILA FIREFOX, MAGENTO, ANDROID, PHP

14.2 Network simulation tool – NS_2

In computer **network**, **network simulation** is a method where a **software program** models the activities of a **network** by calculating the communication between the different **network** objects such as(routers, nodes, switches, access points, links etc.). A **network** simulator is a software program that replicates the functioning of a **computer network**. In simulators, the computer network is typically demonstrated with devices, traffic etc. and the performance is evaluated. Normally, users can then adapt the simulator to accomplish their precise analysis needs. The network parameters that define the state of the network (node placement, existing links) and the events (data transmissions, link failures, etc.). A significant output of simulation is the trace files. Trace files can document every incident that happened in the simulation and are used for examination.

NS2 is the abbreviation of NETWORK SIMULATOR version 2. It was considered explicitly for exploration in network communication and event-driven open-source simulator in computer.

OTCL and c++ are used to create and run NS2. NS2 works on Windows and Linux platforms, that supports wired or wireless network and also use the command line interface as a user interface, API is a pure event base software tool with super simulation design, it has more models which help the user to get desired output easily.

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